

Product Requirements Document (PRD)

XIMB Mess Tracker - Automated Invoice Processing System

Version: 1.0 **Last Updated:** July 2026 **Status:** Production (Live)

1. Problem Statement

The Pain

Imagine you eat lunch and dinner at your school cafeteria every day. Every time you buy something extra (like a cold drink or snacks), they give you a little paper receipt.

At the end of the month: **"How much did you spend?"**

Now you have to:

1. Find 30 emails with receipts
2. Open each PDF receipt one by one
3. Read every single item and price
4. Type it all into a spreadsheet
5. Add everything up
6. Subtract what you already paid

That's exhausting. And boring. And you WILL make mistakes.

The Solution

What if a bot did all of that for you?

That's exactly what XIMB Mess Tracker does. You sign in once with Google, and the robot:

- Opens your email
- Finds the invoice PDFs
- Reads every item and price
- Saves it all
- Shows you a beautiful daily spending diary
- Does this every single day, automatically

You do nothing. The bot does everything.

2. Target Users

USER TYPE	DESCRIPTION	PAIN LEVEL
XIMB Students	~500 students eating at the campus mess daily	High - manual tracking takes 30+ min/month
Mess Committee	Student body managing mess operations	Medium - need aggregate spending data
Individual Trackers	Students who want to budget their food spending	Low - want visibility, not just numbers

User Persona

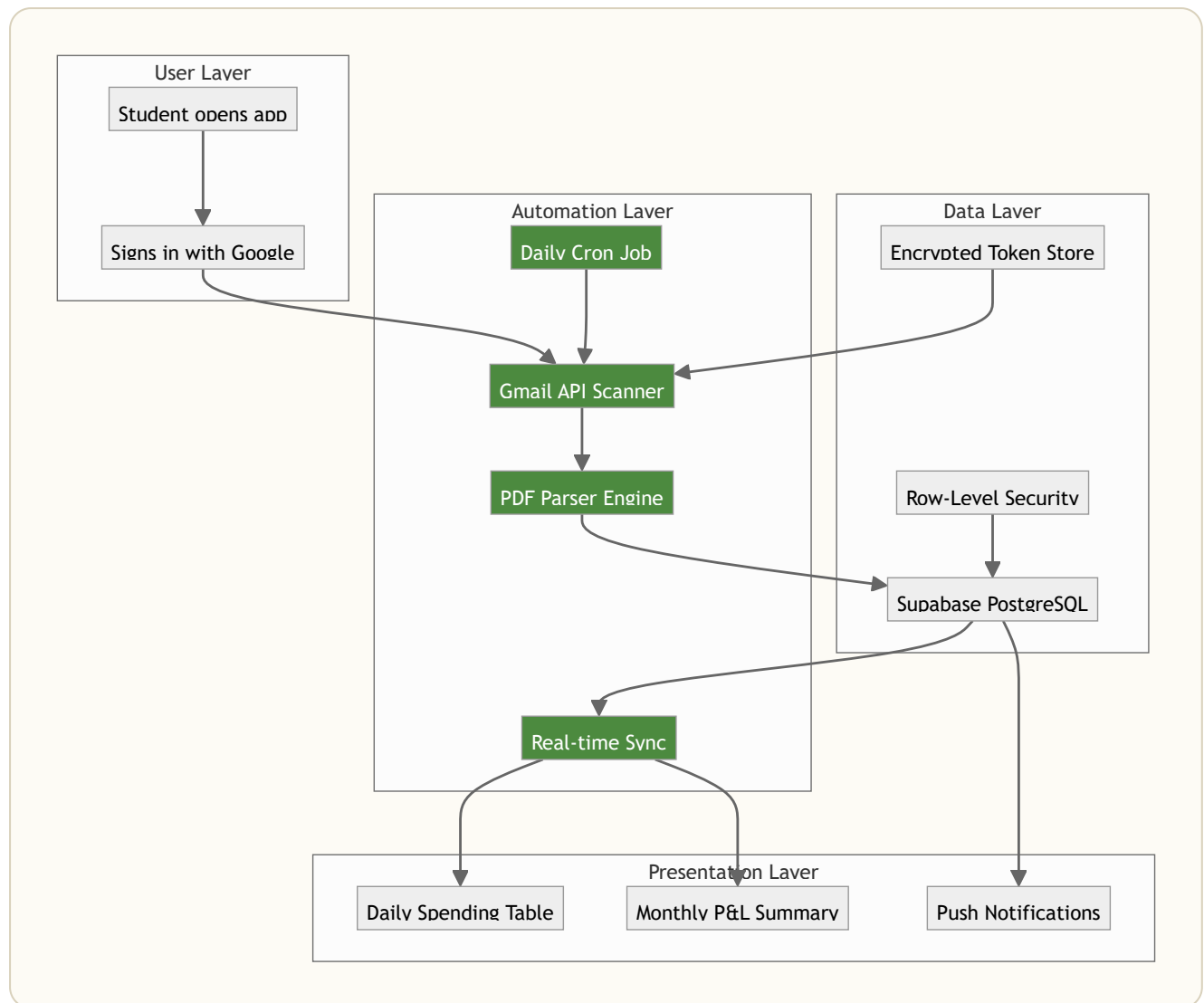
Aditya, 22, MBA Student at XIMB

"I get an invoice email every time I buy something extra at the mess. By month-end, I have 20-30 emails. I used to open each one, note down the amount, and add it up in my head. Half the time I'd forget or get the math wrong. Now I just open the app and everything's already there."

3. Automation Architecture

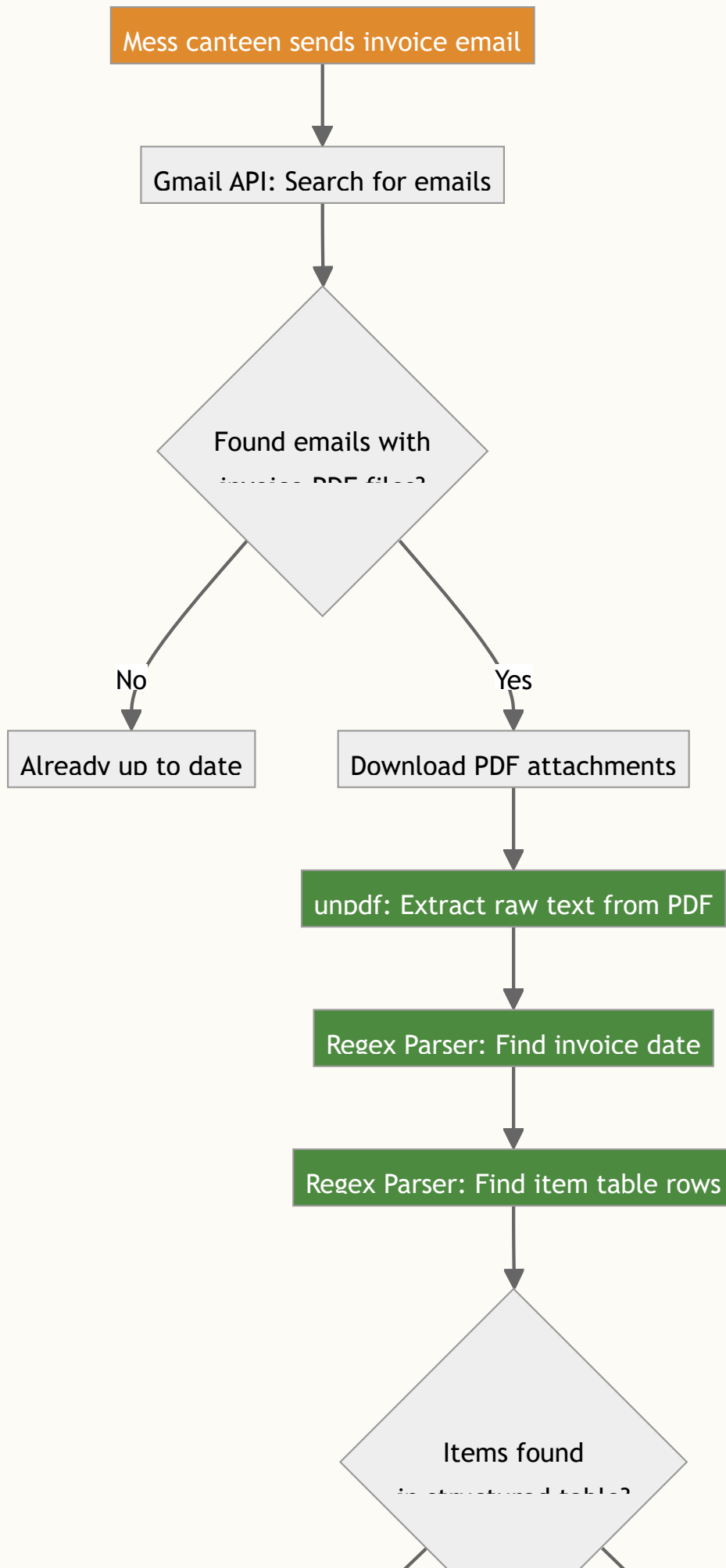
This is the core differentiator. **Everything is automated.** Here's the complete system:

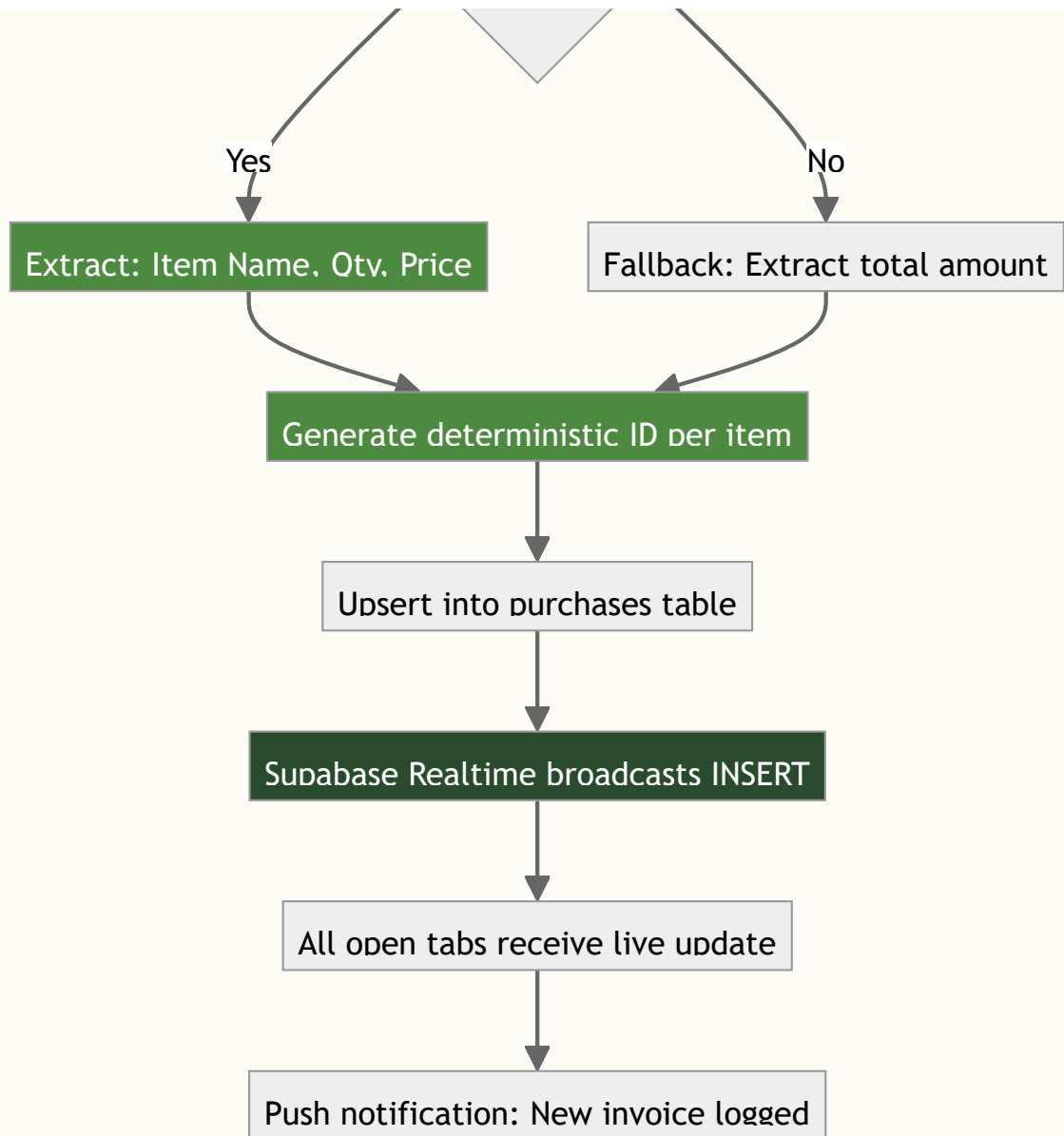
3.1 High-Level System Architecture



3.2 The Automation Pipeline (Detailed Flow)

This is the step-by-step journey of an invoice from email to dashboard:

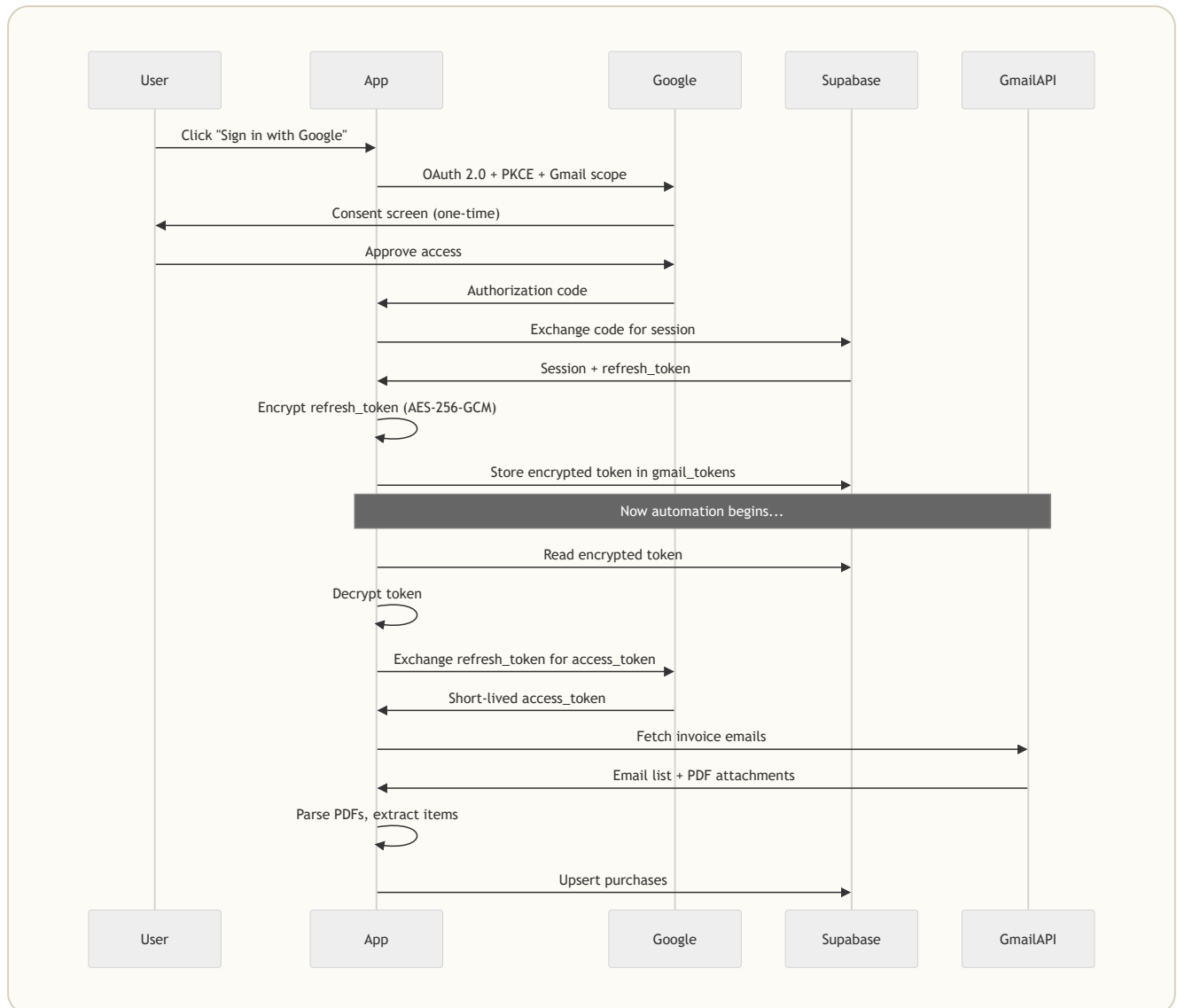




Green boxes = Automated by code (no human involved)

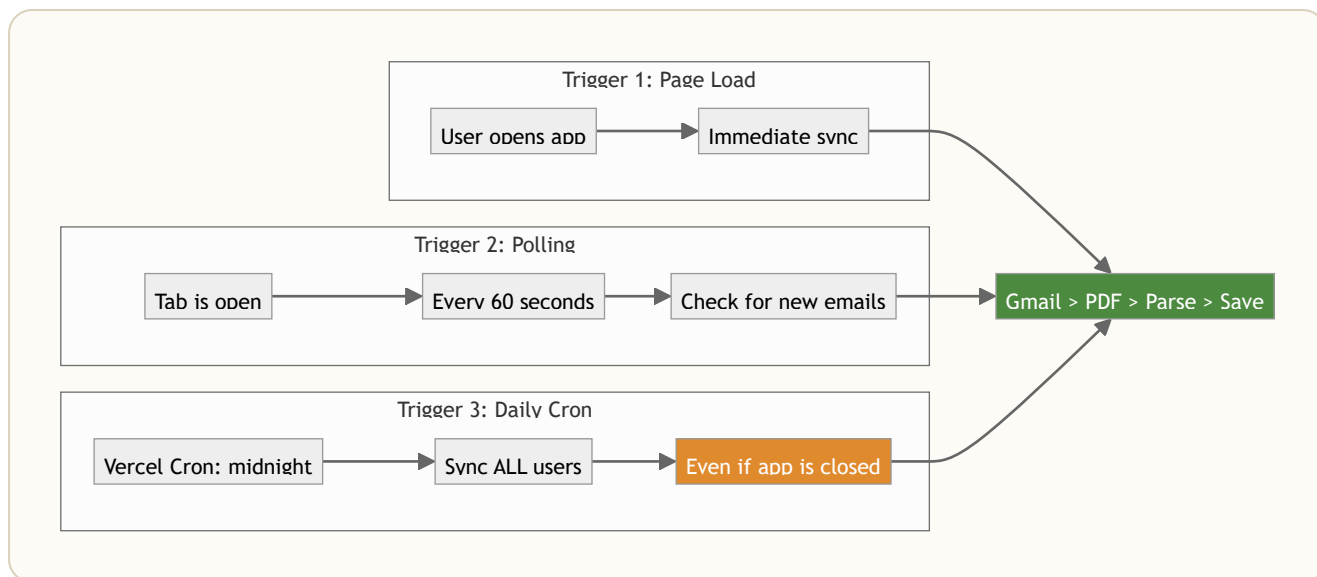
Orange box = The only external trigger (an email arriving)

3.3 Authentication & Security Flow



3.4 Three Sync Triggers

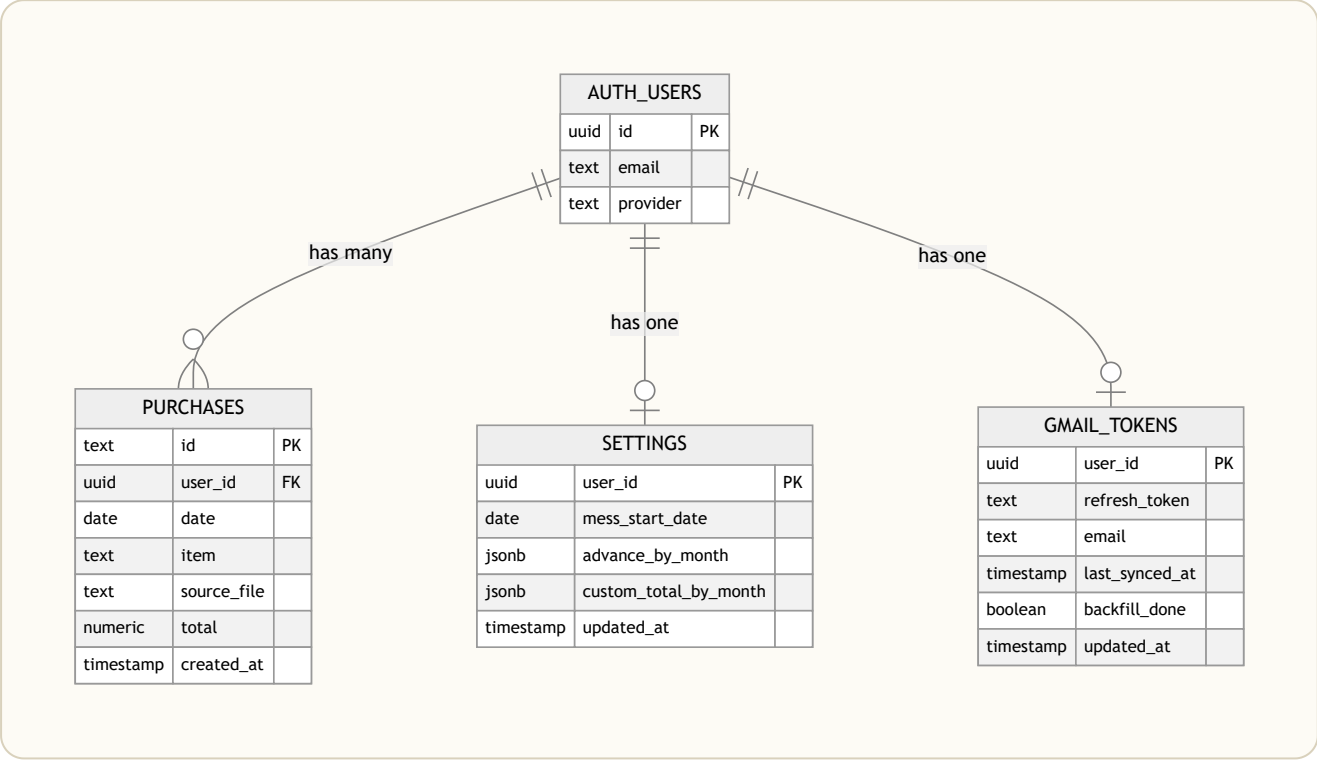
The automation doesn't just run once. It runs through **three different triggers**:



TRIGGER	WHEN	WHO	WHY
Page Load	User opens the app	Single user	Catch up on any missed invoices
60s Polling	While tab is open	Single user	Near-real-time for active users
Daily Cron	Midnight, every day	ALL users	Ensures nobody falls behind, even if they don't open the app

4. Data Model

4.1 Entity Relationship Diagram



4.2 Table Details

purchases - Every food item you bought

COLUMN	TYPE	DESCRIPTION
id	TEXT	Deterministic ID: gm-{userId}-{msgId}-{attachIdx}-{itemIdx} for Gmail items, UUID for manual entries
user_id	UUID	Who bought this
date	DATE	When they bought it
item	TEXT	What they bought (e.g., "Paneer Butter Masala (2)")
source_file	TEXT	Which invoice PDF this came from
total	NUMERIC	How much it cost (Rs)

Why deterministic IDs? So the same invoice processed twice doesn't create duplicate entries. The ID is built from the Gmail message ID + attachment index + item index. This makes the entire pipeline **idempotent**. You can run it 100 times and get the same result.

`gmail_tokens` - Encrypted Gmail credentials

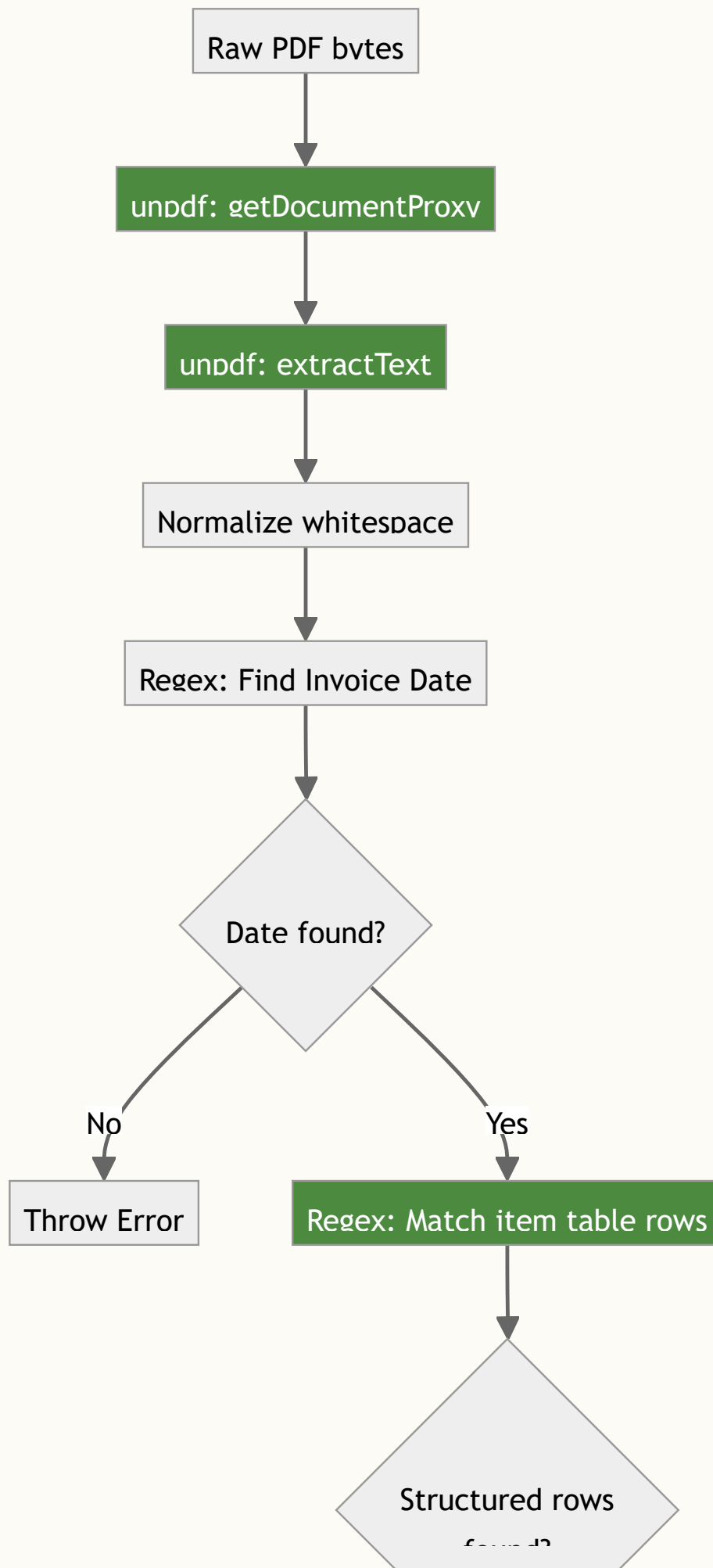
COLUMN	TYPE	DESCRIPTION
<code>user_id</code>	UUID	Token owner
<code>refresh_token</code>	TEXT	AES-256-GCM encrypted Google refresh token
<code>last_synced_at</code>	TIMESTAMPTZ	Watermark for incremental sync
<code>backfill_done</code>	BOOLEAN	Has the full history been scanned?

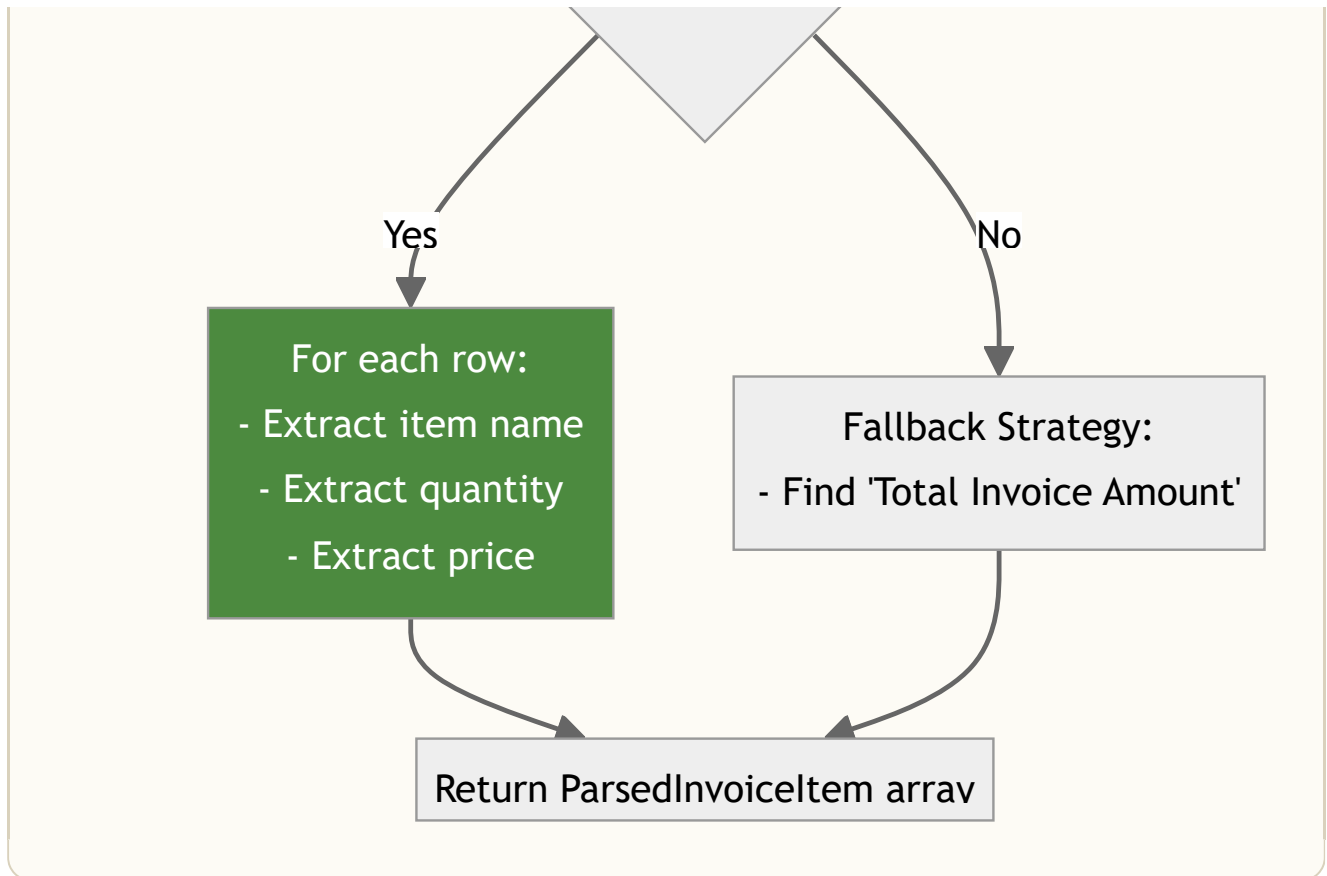
Security: No RLS policies exist on this table. It has RLS enabled but zero policies. This means only the `SUPABASE_SERVICE_ROLE_KEY` (server-side) can read/write tokens. The browser client can never access them.

5. Automation Deep Dive

5.1 PDF Invoice Parsing Engine

The parser is the "brain" of the system. It takes raw PDF bytes and produces structured data.





What the parser understands:

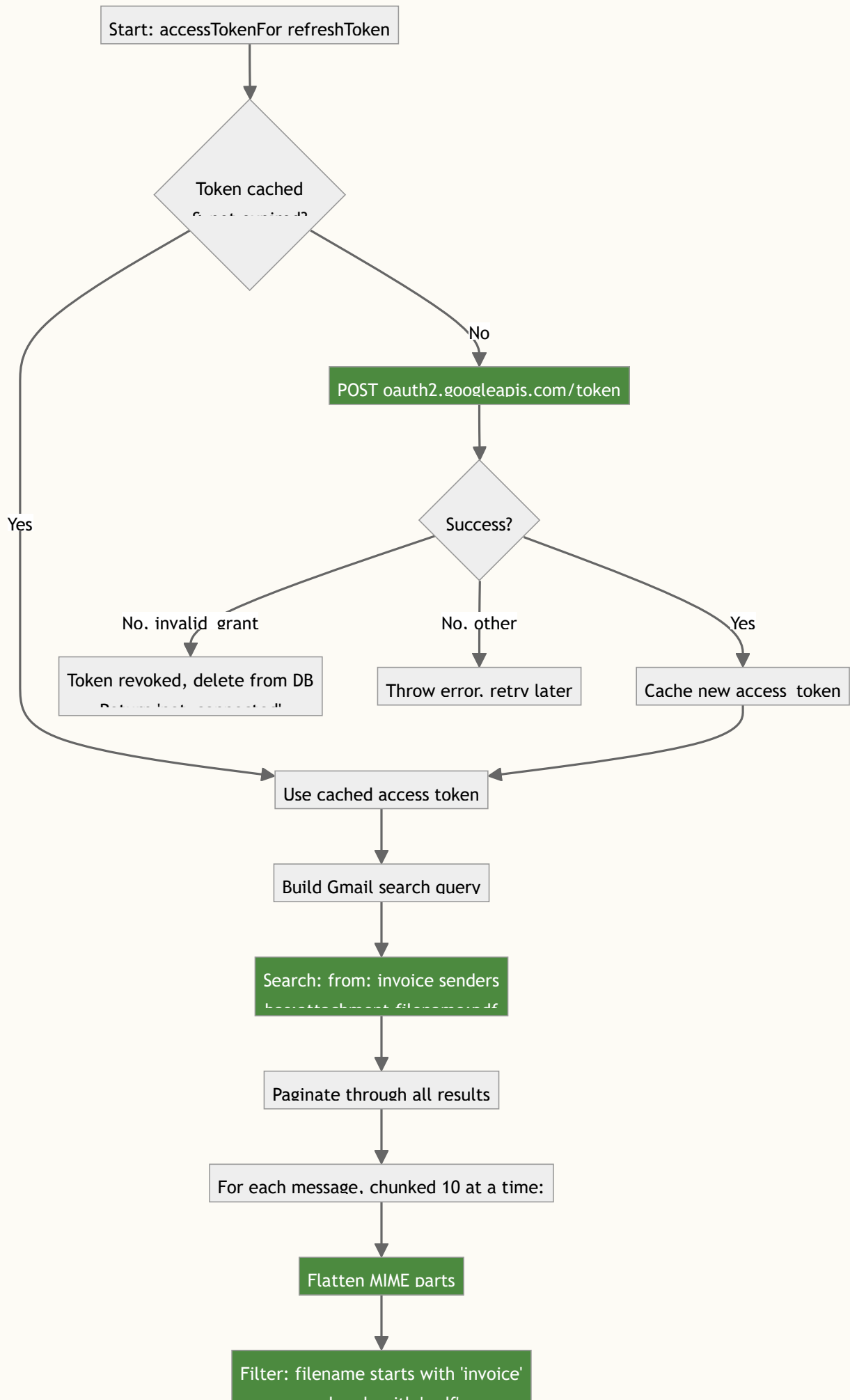
Input PDF text (simplified):

"Invoice Date: 15-07-2026 ... 1 PANEER BUTTER MASALA 2 2.0PC ... 90.00 ... 2 COLD DRINK ..."

Output:

```
[  
  { date: "2026-07-15", item: "Paneer Butter Masala (2)", total: 90.00 },  
  { date: "2026-07-15", item: "Cold Drink", total: 30.00 }  
]
```

5.2 Gmail Scanning Engine





```
graph TD; A[ ] --> B[Download attachment data]; B --> C[Return FetchedInvoice array];
```

Download attachment data

Return FetchedInvoice array

Search query used:

```
from:(noreply@wepsol.com) has:attachment filename:pdf
```

Rate limiting: Messages are processed in chunks of 10 to stay under Gmail's per-user API quota.

5.3 Encryption System

Decrvpt (svnc)

Stored: iv.authTag.ciphertext



Split by dots



Base64 decode each part



AES-256-GCM decrvpt

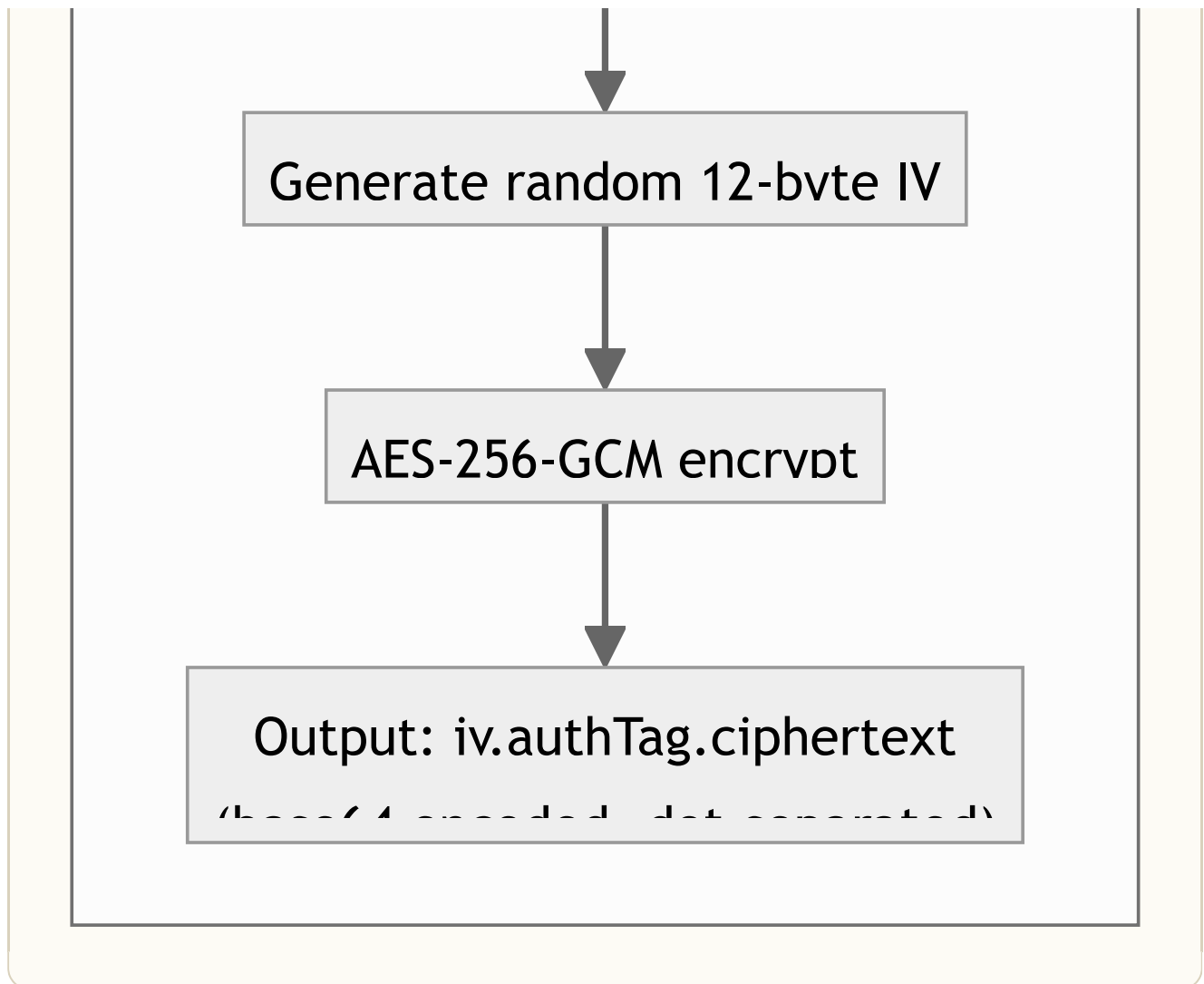


Plain refresh token

Encrvpt (store-token)

Plain refresh token



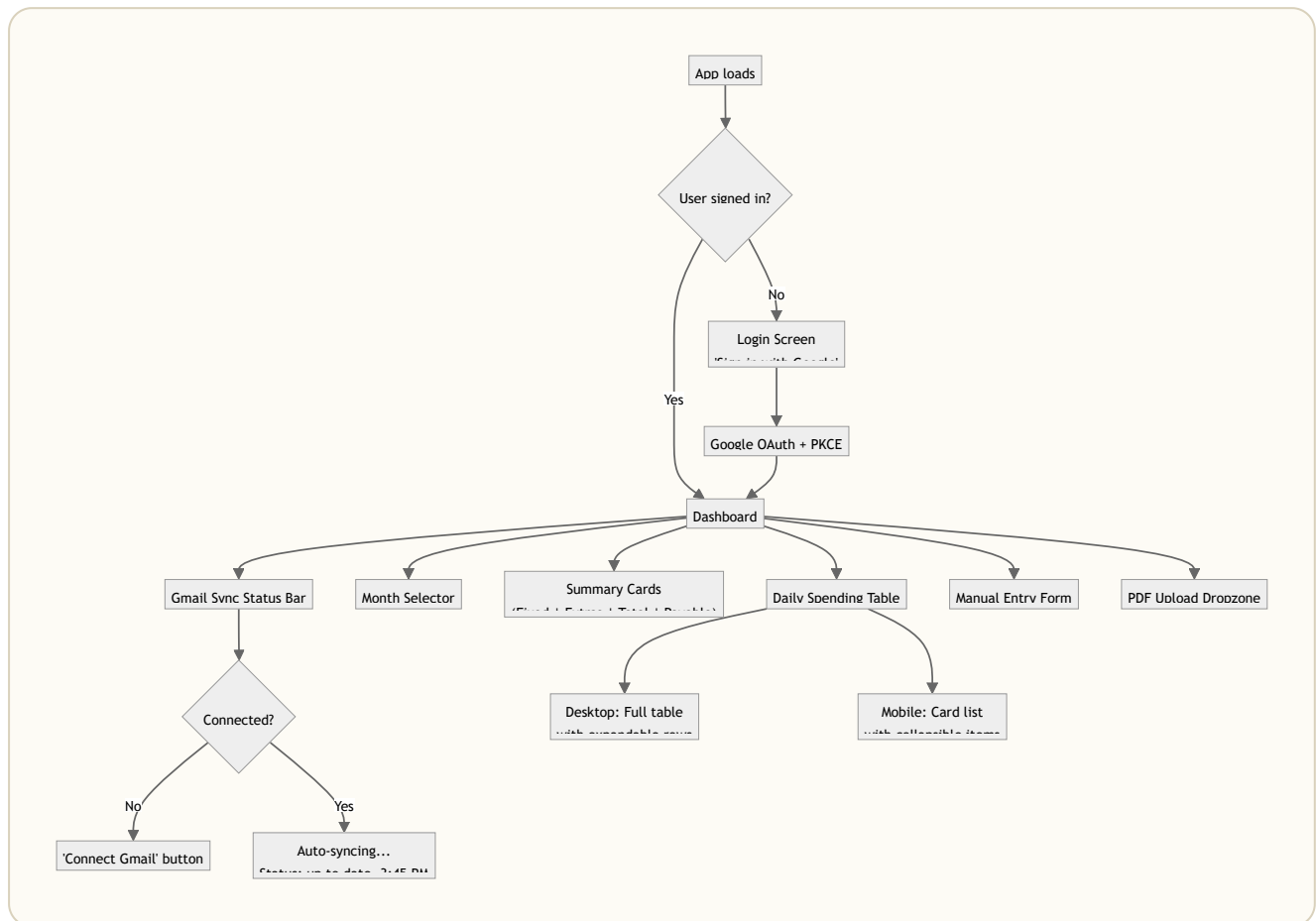


Why AES-256-GCM?

- **AES-256** = Military-grade encryption (even governments use this)
 - **GCM** = Also verifies the data wasn't tampered with (authentication tag)
 - **Random IV** = Same token encrypted twice produces different output (prevents pattern analysis)
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6. User Interface

6.1 Screen Flow

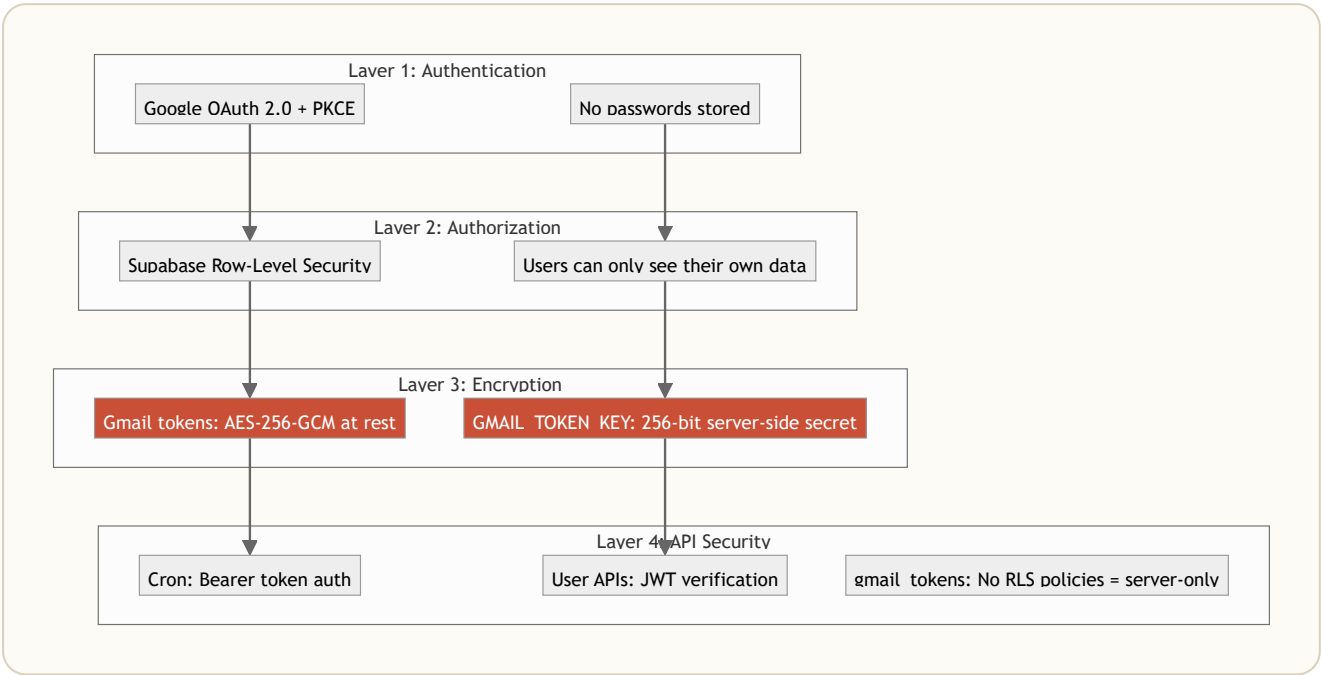


6.2 Key UI Components

COMPONENT	WHAT IT SHOWS	AUTOMATION ROLE
Gmail Sync Bar	Green/red dot, sync status, last sync time	Shows automation is working
Summary Cards	Fixed total, variable total, grand total, payable	Auto-calculated from synced data
Daily Table	Date, fixed cost, extras (expandable), daily total	Auto-populated from parsed invoices
Month Selector	Switch between months	Filters auto-synced data
Payable Bar	Final amount owed (total minus advance)	The "answer", fully automated

7. Security Model

7.1 Defense in Depth



7.2 Key Security Decisions

DECISION	RATIONALE
PKCE auth flow	Prevents authorization code interception attacks (no client_secret in browser)
Encrypted refresh tokens	Even if the database is breached, tokens are useless without the encryption key
Server-only token access	gmail_tokens table has RLS enabled with zero policies = browser JS can never read it
Deterministic purchase IDs	Prevents duplicate entries and makes the pipeline safely re-runnable
1-hour sync overlap window	Ensures no invoices are missed at sync boundaries

8. Automation Configurations

8.1 Vercel Cron Configuration

```
{
  "crons": [
    {
      "path": "/api/gmail/sync-all",
      "schedule": "0 0 * * *"
    }
  ]
}
```

Translation: "Every day at midnight UTC, call the sync-all endpoint to process all connected users."

8.2 Invoice Sender Whitelist

```
export const INVOICE_SENDERS = ["noreply@wepsol.com"];
```

Only emails from `noreply@wepsol.com` (the official canteen automated invoice sender) are scanned. This prevents the app from reading personal or unrelated emails.

8.3 Attachment Filter Rules

Accepted: invoice-july-2026.pdf, Invoice_15072026.pdf
Rejected: receipt.pdf, menu.pdf, photo.jpg

Rule: Filename must **start with "invoice"** AND **end with ".pdf"**.

9. Performance & Scalability

Metric	Current	Design Limit
Users	3 active	Unlimited (each syncs independently)
Sync latency	~2-5 seconds per user	60-second max duration (Vercel)

METRIC	CURRENT	DESIGN LIMIT
PDF parse time	~200ms per invoice	Serverless scales horizontally
Gmail API calls	Chunked (10 messages/batch)	Stays under per-user quota
Cron frequency	Daily (midnight UTC)	Can increase to hourly if needed
Real-time updates	~100ms (Supabase Realtime)	WebSocket, no polling

10. Future Roadmap

PRIORITY	FEATURE	AUTOMATION LEVEL
P0	Multi-canteen support	Auto-detect canteen from invoice format
P1	Spending insights	Auto-generated weekly spending summaries
P1	Budget alerts	Auto-notify when spending exceeds threshold
P2	WhatsApp sync	Parse invoices from WhatsApp messages
P2	Receipt OCR	Camera capture to OCR to auto-log (for paper receipts)
P3	Mess committee dashboard	Aggregate analytics for mess management

11. Success Metrics

METRIC	TARGET	HOW WE MEASURE
Zero-touch rate	>90% of purchases auto-logged	<code>source_file LIKE 'invoice%'</code> / total purchases
Sync reliability	>99% daily cron success	Vercel Cron logs
Time saved	30+ min/month per user	User survey
Parse accuracy	>95% items correctly extracted	Manual spot-checks
Active users	50+ monthly	PostHog analytics

12. Testing Strategy

LAYER	WHAT	HOW
Parser	Invoice PDF extraction accuracy	Sample PDFs with known items, compare output
Sync	End-to-end Gmail to DB pipeline	Test accounts with known invoice emails
Auth	OAuth flow + token storage	Manual browser testing
Security	Token encryption/decryption roundtrip	Unit test: encrypt then decrypt = original
Cron	Daily sync-all reliability	Vercel Cron monitoring dashboard

XIMB Mess Tracker. Because life's too short to count mess bills by hand.

Automated. Zero effort.